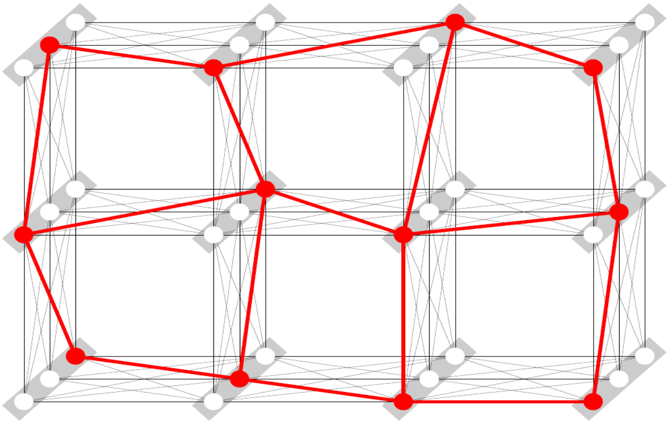
Combinatorial Solution Set

$$x^* = \min_{\overline{x} \in X_V} E_\theta(\overline{x}) := \min_{\overline{x} \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$



Number of possible labelings: $\prod_{v \in V} |X_V|$

Example: 4x4 grid and full graphs with 10 labels

1) 400
2) 440
3) 40
4) 1000
5) 10000
6) No correct answer

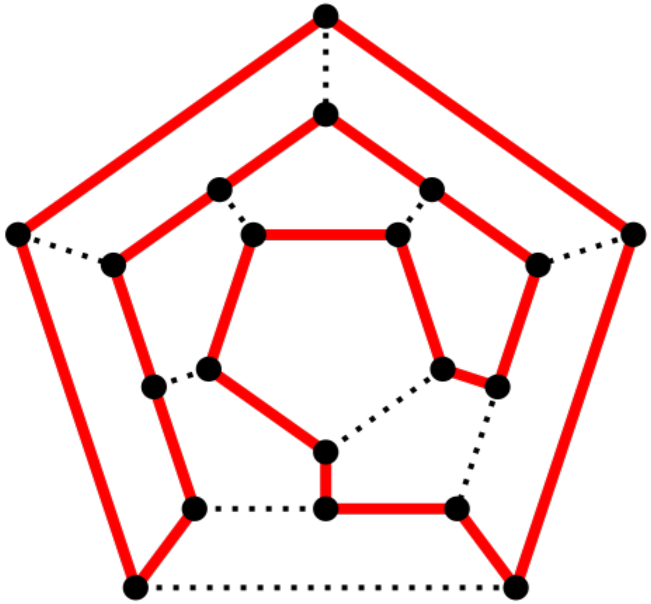
Picture: T Werner. A Linear Programming Approach to Max-sum Problem: A Review

Combinatorial Solution Set = No Polynomial Algorithm?

1)

Hamiltonian Cycle

A **Hamiltonian cycle** in a undirected graph graph is the cycle that visits each vertex exactly once.



Determining whether such a cycle exists in a graph is NP-complete.

Picture: Wikipedia

Hamiltonian Cycle Reduces to MAP-Inference

Graph $(V, \mathcal{E})$ for a Hamiltonian cycle

Full graph $(V, V \times V)$ for the MAP-inference problem

The set of labels $X_v = V$ for all $v \in V$.

What about an inverse reduction:
MAP-inference to a Hamiltonian cycle?

How to Deal With NP-Hard Problems?

1) Identify polynomially solvable subclasses

2) Consider its convex relaxations (polynomially solvable approximations)

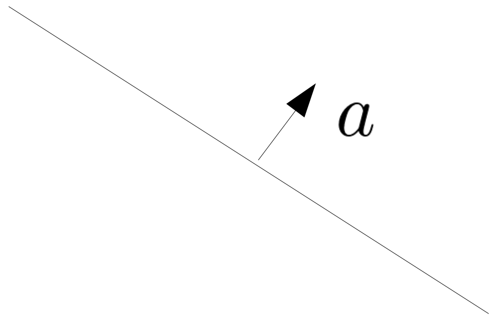
3) Develop approximate algorithms

We start with some background...

Linear Programs and Their Geometry

Integer linear programs is a standard way to describe combinatorial problems.

Hyperplane: $\{x \in \mathbb{R}^n \colon \langle a, x \rangle = b\}$



Half-space $\{x \in \mathbb{R}^n \colon \langle a, x \rangle \leq b\}$

Polyhedron: $P = \{x \in \mathbb{R}^n \colon Ax \leq b\}$

Polytope:

Linear Programs and Their Geometry

Polyhedron: $Ax \leq b$

Polyhedron?

$$Ax = b$$
$$x_i \geq 0, \; i \in I \subset \{1, \ldots, n\}$$

1) Yes
2) No

Linear Programs and Their Geometry

Polyhedron: $Ax \leq b$

Polyhedron?

$$Ax = b$$
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1) Yes
2) No

$$\mathrm{Ax} \leq b$$
$$-Ax \leq -b$$
$$-x_i \leq 0, i \in I$$

Linear Programs and Their Geometry

What about an inverse transformation:

$$Ax \leq b \quad \text{to} \quad \begin{array}{l} Ax = b \\ x_i \geq 0,\ i \in I \subset \{1, \ldots, n\} \end{array} \quad ?$$

Linear Programs and Their Geometry

What about an inverse transformation:

$Ax \leq b$ to $\begin{aligned} &Ax = b \\ &x_i \geq 0,\ i \in I \subset \{1, \dots, n\} \end{aligned}$?

$$
\begin{aligned}
Ax + x' &= b \\
x' &\geq 0
\end{aligned}
$$

Linear Programs and Their Geometry

Linear programs (also holds for **max**):

Standard form

$$\min_{x \in \mathbb{R}^n} \langle c, x \rangle$$

$$Ax \leq b$$

Canonical form

$$\min_{x \in \mathbb{R}^n} \langle c, x \rangle$$

$$Ax = b$$

$$x_i \geq 0, \ i \in I \subset \{1, \ldots, n\}$$

Vertexes and solutions:

Linear Programs and Their Geometry

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Vertexes and solutions:

$x' \in \mathbb{R}^n$ is a vertex of a polyhedron P if there exists $c \in \mathbb{R}^n$ such that

$$\max_{x \in P} \langle c, x \rangle$$

is finite and has a unique solution

N-Dimensional Simplex

$$\Delta^n = \{p \in \mathbb{R}^n_+ : \sum_{i=1}^{n} p_i = 1\}$$

Examples: 1) Yes/ 2) No

0.3; 0.1; 0.7; 0.0;

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N-Dimensional Simplex

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Draw your own picture:

Convex Hull

$$\delta^i \in \mathbb{R}^N, \; i = 1, \ldots, n$$ - arbitrary vectors

Convex hull: **Always a polytope!**

$$\mathrm{conv}\{\delta^i \colon i = 1, \ldots, n\} := \{s \colon s = \sum_{i=1}^{n} p_i \delta^i, \; p \in \Delta^n\}$$

Picture:

Convex Hull

Theorem 1:

Polyhedron *P* is a polytope if and only if it is representable as a convex hull of its vertexes.

Theorem 2:

Let $\delta \in \mathbb{R}^N$ be a vertex of $\mathrm{conv}\{\delta^i : i = 1, \ldots, n\}$

Then there exists $i \in \{1, \ldots, n\}$ such that $\delta = \delta^i$.

Convex Hull

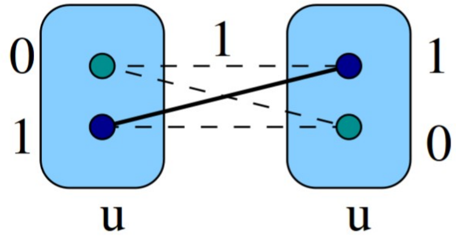
Lemma 1:

$$\min_{i=1,\dots,n}\{a_i\} = \min_{p\in\Delta^n}\sum_{i=1}^{n} p_i a_i = \min_{\mu\in\mathrm{conv}\{a_i,\ i=1,\dots,n\}} \mu$$

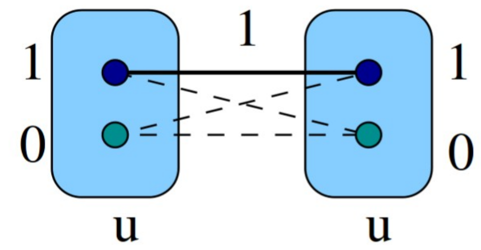
Marginal Polytope: Labeling → Vector

Mapping: labeling → binary vector

$$\delta : X_V \to \{0,1\}^I$$



$$\delta(x^1) = \underbrace{01}_{u}\ \underbrace{0100}_{uv}\ \underbrace{10}_{v}$$

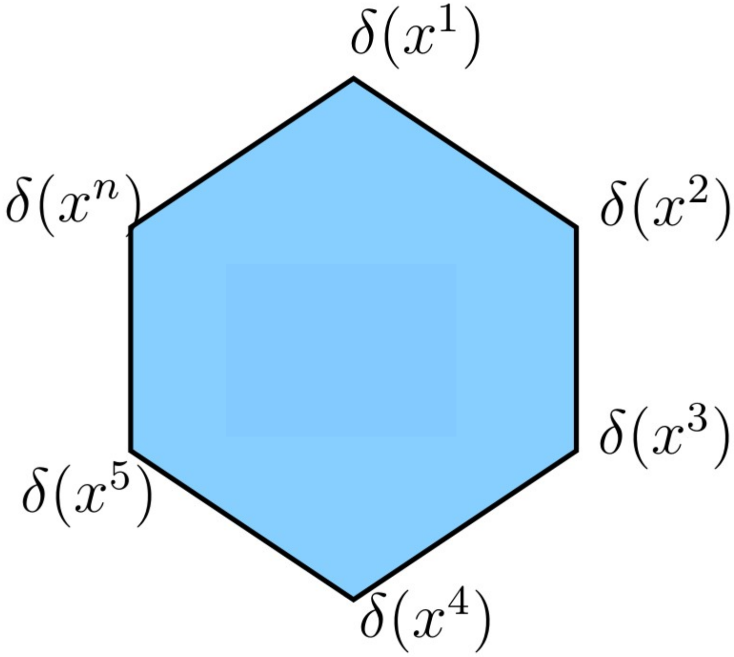


$$\delta(x^2) = \underbrace{10}_{u}\ \underbrace{1000}_{uv}\ \underbrace{10}_{v}$$

$$\min_{x \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v) \equiv \min_{x \in X_V} \langle \theta, \delta(x) \rangle$$

Marginal Polytope

Marginal polytope: $M := \mathrm{conv}\{\delta(x) \in \mathbb{R}^I : x \in X_V\}$



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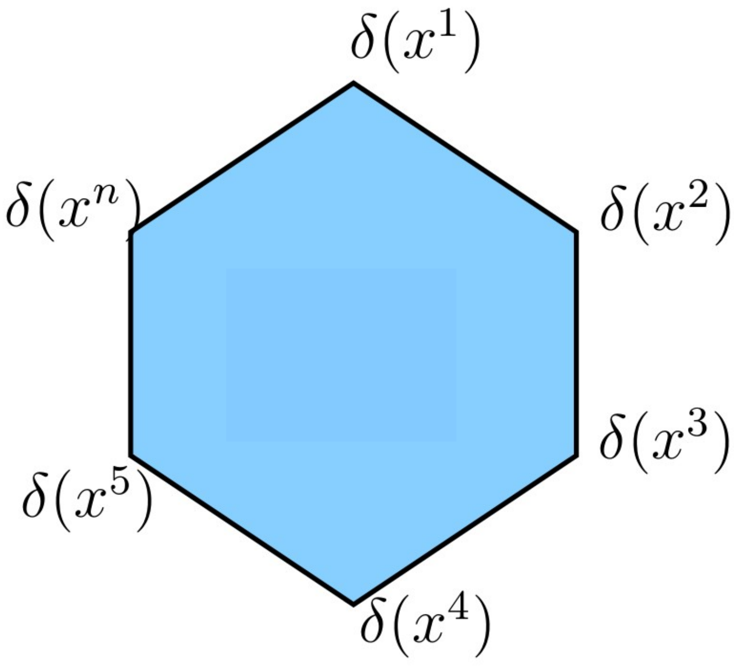
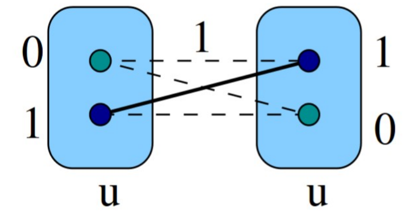
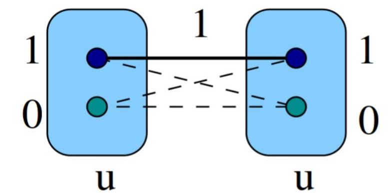


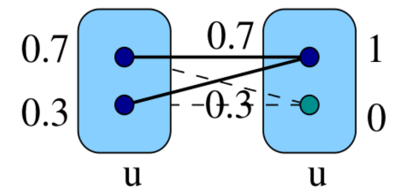


$$\delta(x^1) = \underbrace{01}_{u}\ \underbrace{0100}_{uv}\ \underbrace{10}_{v}$$



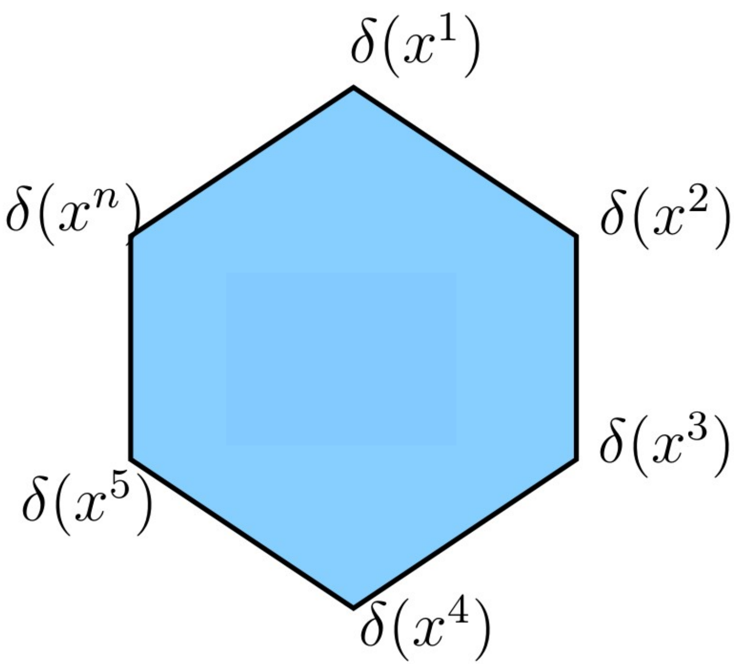
$$\delta(x^2) = \underbrace{10}_{u}\ \underbrace{1000}_{uv}\ \underbrace{10}_{v}$$

$0.3x^1 + 0.7x^2$:



$$0.3\delta(x^1) + 0.7\delta(x^2) = \underbrace{0.7\,0.3}_{u}\ \underbrace{0.7\,0.3\,0\,0}_{uv}\ \underbrace{10}_{v}$$

Marginal polytope: $$M := \mathrm{conv}\{\delta(x) \in \mathbb{R}^I : x \in X_V\}$$

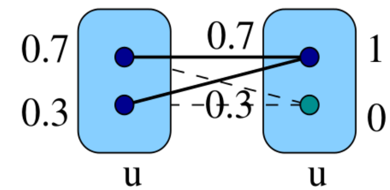


Corollary 1:

$$\mu_v \in \Delta^{X_v} \text{ for any } v \in V$$

$$\mu_{uv} \in \Delta^{X_{uv}} \text{ for any } uv \in \mathcal{E}$$

$0.3x^1 + 0.7x^2$:



$$0.3\delta(x^1) + 0.7\delta(x^2) = \underbrace{0.7\ 0.3}_{u}\,\underbrace{0.7\ 0.3\ 0\ 0}_{uv}\,\underbrace{1\ 0}_{v}$$

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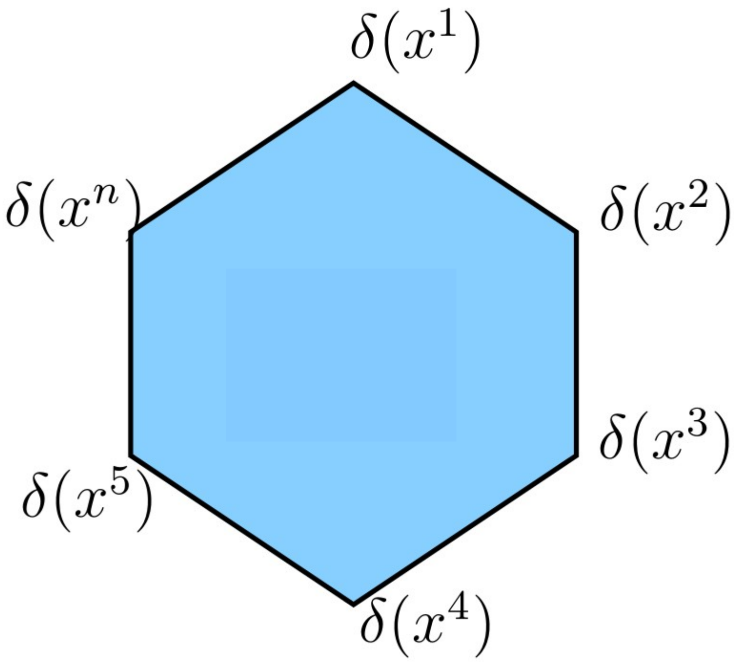

Proposition 2:

$$\min_{x \in X_V} \langle \theta, \delta(x) \rangle = \min_{\mu \in M} \langle \theta, \mu \rangle$$

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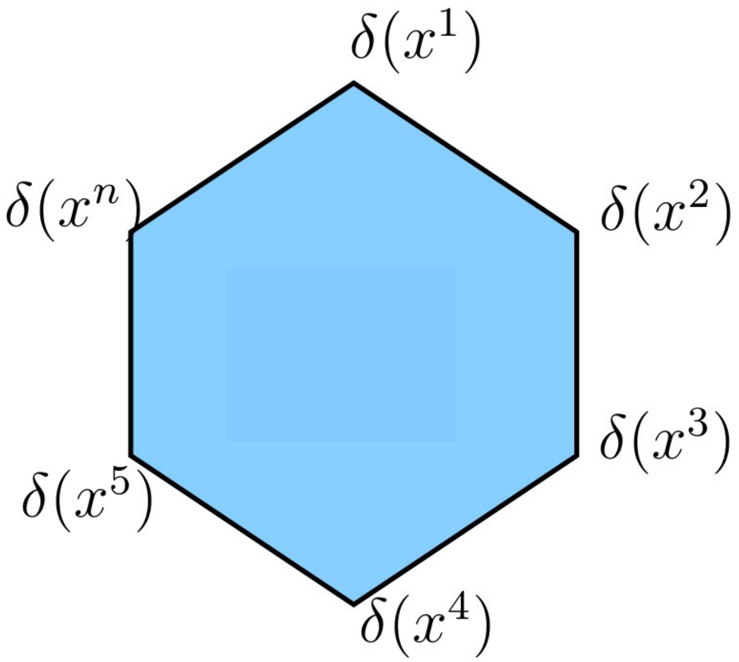

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NP-hard problem = linear problem?

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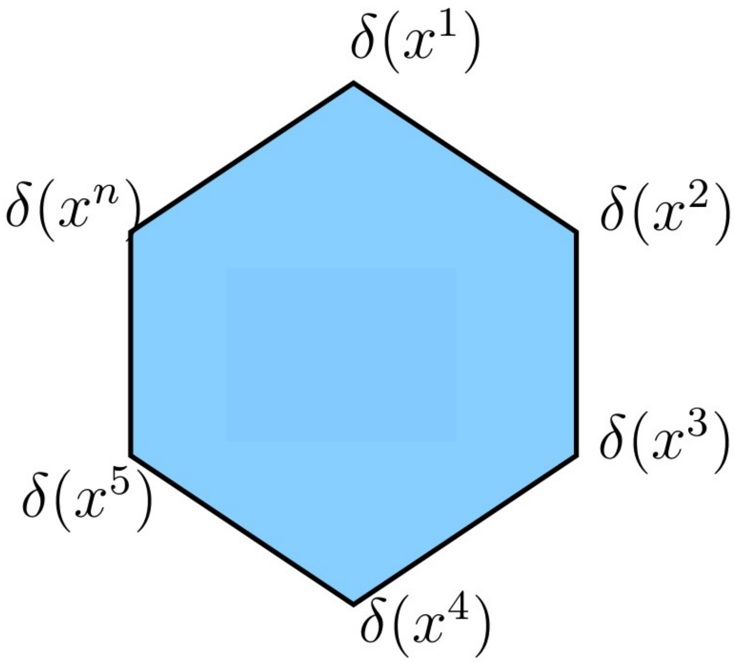

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NP-hard problem = linear problem?

$$\sum_{x \in X_V} p_x \delta(x) = \mu$$

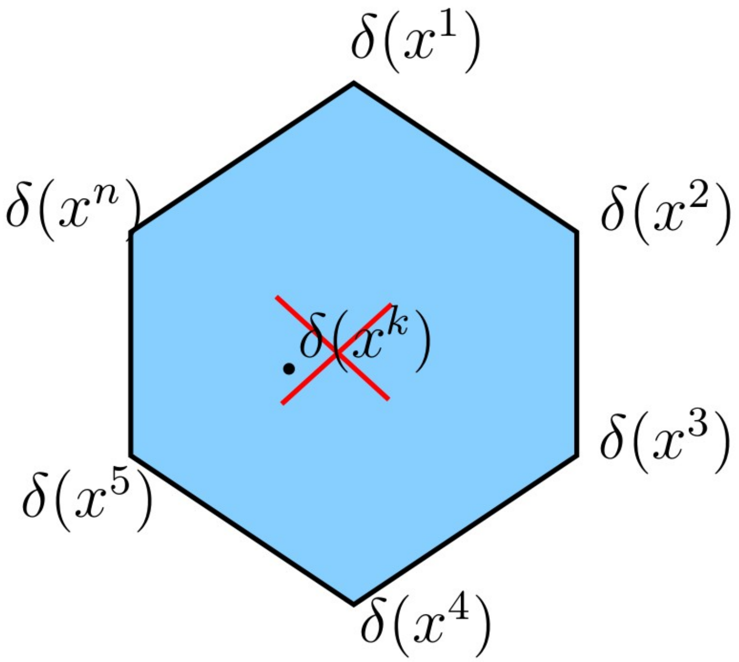
$$\sum_{x \in X_V} p_x = 1$$

$$p_x \geq 0, \quad x \in X_V$$

Yes, with an exponential number of constraints!

Marginal Polytope

Marginal polytope: $M := \text{conv}\{\delta(x) \in \mathbb{R}^I : x \in X_V\}$



Proposition 3:

For any $x \in X_V$

the vector $\delta(x)$

is a vertex of M.